Chunsu Park

https://chunsupnu.github.io/

Interests

Image Science, Information Theory, Medical Imaging, Task-informed Imaging and Assessment, Generalization, Generative AI, Explainable AI

Education

University of Illinois at Urbana-Champaign Aug. 2025 – present
Ph.D. student in Bioengineering Champaign, United States

Pusan National University Mar. 2021 – Feb. 2023
Master of Science in Information Convergence Engineering Busan, Korea

- Graduation Thesis: "Deep learning networks and interpretation for bone marrow edema detection in dual-energy CT" (Major: Artificial Intelligence)
- GPA: 4.44/4.5 (4.0 scale: 4.0)

Jeonbuk National University Mar. 2010 – Feb. 2017
Bachelor of Science in Industrial and Information Systems Engineering Jeonju, Korea
Bachelor of Business Administration

- Graduated as the *Salutatorian* in the department (2/46)
- Military Service in Republic of Korea Army during 2011 – 2013
- GPA: 4.05/4.5 (4.0 scale: 3.64)

Selected Publications

(†:co-first, *:corresponding)

H. Lee†, <u>Chunsu Park</u>†, M.W. Kim, C. Park*, "Deep Learning-Based Artifact Reduction: Radiologist and AI Classifier Evaluation of Dual-Energy CT Image Quality in Femoral Bone Marrow Edema," *European Journal of Radiology Artificial Intelligence*, 2025, 100062.

<u>Chunsu Park</u>†, H. Lee†, M.W. Kim, C. Park*, "Deep learning-based Segmentation in Musculoskeletal Imaging: A Review of Research Trends," *Journal of the Korean Society of Radiology*, 2025, 86 (5), 587.

S. Lee, S. Kim, M. Seo, S.K. Park, S. Imrus, A. Kambaluru, D.E. Lee, **<u>Chunsu Park</u>**, S.Y. Lee, J.Kim, J.H. Yoo, M.W. Kim*, "Enhancing Free-hand 3D Photoacoustic and Ultrasound Reconstruction using Deep Learning," *IEEE Transactions on Medical Imaging (IF: 9.8, JCR 2024 <3%)*, 2025.

<u>Chunsu Park</u>, S. Kim, D.E. Lee, S.Y. Lee, A. Kambaluru, C. Park, M.W. Kim*, "CAPTURE-GAN: Conditional Attribute Preservation through Unveiling Realistic GAN for artifact removal in dual-energy CT imaging," *The 27th Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2024, Marrakesh, Morocco. (857/2879 ≈ 29.8%)[Link] [Code]

D.E. Lee, **<u>Chunsu Park</u>**, S.Y. Lee[1], S.Y. Lee[2], M.W. Kim*, "Convolutional Implicit Neural Representation of pathology whole-slide images," *The 27th Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2024, Marrakesh, Morocco. (857/2879 ≈ 29.8%) [Link] [Code]

<u>Chunsu Park</u>, J.W. Kang, D.E. Lee, W. Son, S.M. Lee, C. Park*, M.W. Kim*, "W-DRAG: A joint framework of WGAN with data random augmentation optimized for generative networks for bone marrow edema detection in dual energy CT," *Computerized Medical Imaging and Graphics (IF: 5.4, JCR 2023 <9%)*, 2024, 115, 102387. [Link] [Code]

J.W. Kang, **<u>Chunsu Park</u>**, D.E. Lee, J.H. Yoo, M.W. Kim*, "Prediction of bone mineral density in CT using deep learning with explainability," *Frontiers in Physiology (IF: 3.2, Q2)*, 2023, 13, 1061911. [Link]

<u>Chunsu Park</u>†, M.W. Kim†, C. Park*, W. Son, S.M. Lee, H.S. Jeong, J.W. Kang, M.H. Choi, "Diagnostic performance for detecting bone marrow edema of the hip on dual-energy CT: Deep learning model vs. musculoskeletal physicians and radiologists," *European Journal of Radiology (IF: 3.2, Q1)*, 2022, 152, 110337. [Link]

Conference Presentations

(*:corresponding)

<u>Chunsu Park</u>, S. Kim, D.E. Lee, S.Y. Lee, A. Kambaluru, C. Park, M.W. Kim*, "A Conditional GAN Approach for Artifact Removal: Preserving Pathological Patterns in Dual-energy CT Imaging," *The 110th Radiological Society of North America (RSNA)*, 2024, Chicago, USA (Scientific Poster Session)

<u>Chunsu Park</u>, D.E. Lee, S. Kim, S.Y. Lee, M.W. Kim*, "Mask CycleGAN for removing artifacts in dual-energy CT," *The 8th IEEE International Conference on Consumer Electronics-Asia (ICCE-Asia)*, 2023, Busan, South Korea (Oral Session)

<u>Chunsu Park</u>, W. Son, H.S. Jeong, S.M. Lee, M.W. Kim, C. Park*, "Diagnostic performance for detecting bone marrow edema on dual-energy CT with Deep learning networks," *The 20th Asian Oceanian Congress of Radiology in conjunction with the 78th Annual Meeting of the Korean Society of Radiology (AOCR & KCR)*, 2022, Seoul, South Korea (Oral Session)

<u>Chunsu Park</u>, D.E. Lee, C. Park, M.W. Kim*, "Deep learning-based bone marrow edema detection using dual-energy CT with multi-channel data fusion," *International Biomedical Engineering Conference (IBEC)*, 2021, Virtual Conference (Oral Session)

Technical Skills

Advanced Python, PyTorch, TensorFlow, LaTeX
Moderate C, MATLAB, MySQL

References

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